Build your custom network topology in Mininet

Shown below is the code for custom topology creation. The first section imports an API for various functions.

```
#/usr/bin/python
from mininet.net import Mininet
from mininet.node import Controller
from mininet.node import RemoteController
from mininet.cli import CLI
from mininet.link import TCLink
from mininet.log import setLogLevel, info
```

Note: In the follwing code, we define a function myNet() which creates all the nodes and links. The Mininet class returns an empty network object to which we add hosts, switches and a controller. The RemoteController class returns a controller object which runs outside Mininet. Then we assign the controller into our Net object.

```
def myNet():
    "Create an empty network"
    net = Mininet( link=TCLink )

   "Create a RemoteController and add to network"
    info( '*** Adding RemoteController\n' )
    c = RemoteController('c', ip='127.0.0.1', port=6666)
net.controllers = [ c ]
Note: net.addHost() creates host nodes and adds to the
network. net.addSwitch adds the openflow switch to the network

  "Create hosts and add to network"
  info( '*** Adding hosts\n' )
  h1 = net.addHost( 'h1', ip='10.0.0.1' )
  h2 = net.addHost( 'h2', ip='10.0.0.2' )
  h3 = net.addHost( 'h3', ip='10.0.0.3' )
  "Create switch and to network"
  info( '*** Adding switch\n' )
  s3 = net.addSwitch( 's3' )
  s4 = net.addSwitch( 's4' )
```

Note: In the following code, we connect the nodes by adding links between them as shown. Then the net.build() call will build the topology and net.start() will start all the nodes. CLI(net) will run the *cli* mode awaiting user inputs.

```
# Add link:
    h1.linkTo( s3 )
    h2.linkTo( s3 )
    s3.linkTo( s4 )
   h3.linkTo( s4 )
# Network build and start:
 net.build()
 net.start()
#CLI mode running:
```

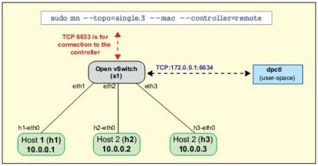

Figure 2: The graphical depiction of the basic network topology

Figure 3: Simulation of the basic topology

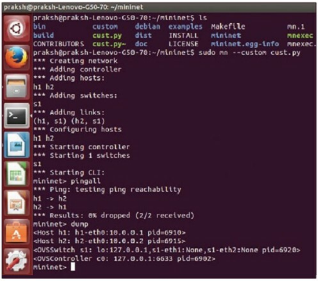

Figure 4: Simulation of custom topology

```
CLI( net )
 net.stop()
if __name__ == '__main__':
 setLogLevel( 'info' )
 myNet()
```

Simulating a basic Mininet network topology using the Mininet command

The following example creates a network topology with a single OpenFlow switch and three hosts connected to a remote controller. When we run this topology, it shows that it's unable to connect to the remote controller. After running

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